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PHYSICS

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Medium Eng./Hindi	ENG	Registration Number	46104254
Center	ONLINE	Date	29/JUNE/26

INDEX TABLE			INSTRUCTIONS	
	Maximum Marks	Marks Obtained	1. Do furnish the appropriate details in the answer sheet (viz. Name, Registration Number and Test Code). 2. Question No. 1 is COMPULSORY. Of the remaining four questions (Q2 to Q5), any THREE are to be attempted. Thus a candidate has to attempt FOUR questions in all, totalling 200 marks. 3. The number of marks carried by a question/part is indicated against it. 4. Answers must be written in the medium authorized in the Admission Certificate, which must be stated clearly on the cover of this Question-Cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in medium other than the authorized one. 5. Word limit in questions, if specified, should be adhered to. 6. Any page or portion of the page left blank in the Question-Cum-Answer Booklet must be clearly struck off.	
Total Marks Obtained:				
Remarks:				
			Is student recommended for One-to-One mentoring?	
			Recommended	Strongly Recommended

16-B, 2nd Floor, Above National Trust Building, Bada Bazar Marg, Old Rajinder Nagar, Delhi-110060

Plot No. 857, 1st Floor, Banda Bahadur Marg (Opp. Punjab & Sind Bank), Dr. Mukherjee Nagar, Delhi-110009

EVALUATION INDICATORS

1. Contextual Competence
2. Content Competence
3. Language Competence
4. Introduction Competence
5. Structure - Presentation Competence
6. Conclusion Competence

Overall Macro Comments / feedback / suggestions on Answer Booklet:

1.

2.

3.

4.

5.

6.

All the Best

भौतिकी / PHYSICS

Mechanics of Particles

Time Allowed: Two Hours

Maximum Marks: 150

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions:

There are **Four** questions printed in **ENGLISH**.

Candidate has to attempt **Three** questions in all.

Questions no. 1 is compulsory and out of the remaining, any **two** are to be attempted.

The number of marks carried by a question / part is indicated against it.

Answers must be written in the medium authorized in the Admission Certificate which must be stated clearly on the cover of this Question-cum-Answer (QCA) Booklet in the space provided. No marks will be given for answers written in a medium other than the authorized one.

Assume suitable data, if considered necessary, and indicate the same clearly.

Unless and otherwise indicated, symbols and notations carry their usual standard meanings.

Attempts of questions shall be counted in sequential order. Unless struck off, attempt of a question shall be counted even if attempted partly. Any page or portion of the page left blank in the Question-cum-Answer (QCA) Booklet must be clearly struck off.

Physical Constants:

Velocity of light

$$c = 3 \times 10^8 \text{ m/s}$$

Boltzmann constant

$$k = 1.38 \times 10^{-23} \text{ J/K}$$

Planck's constant

$$h = 6.627 \times 10^{-34} \text{ J s}$$

Permittivity of free space

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

Permeability of free space

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

Charge of the electron

$$e = 1.6 \times 10^{-19} \text{ C}$$

Mass of the electron

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

Universal gas constant

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

Universal gravitational constant

$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

Q1. (a) For a system of N particles show that angular momentum is conserved in the absence of external torque. 10

(b) Calculate the time period of revolution of International Space Station. Assume a circular orbit of 418 km above earth's surface. (Radius of earth = 6,380 km) 10

(c) The eccentricity of earth's orbit is 0.0167. Calculate the ratio of maximum and minimum speeds of earth in its orbit. 10

(d) Prove that work done by a non-conservative force on a particle is change in its mechanical energy. 10

(e) Define a conservative force field. Determine if the force field given below is conservative in nature. 10

$F = (y^2 * z^3 - 6x * z^2)i + (2x * y * z^3)j + (3x * y^2 * z^2 - 6x^2 * z)k$. i, j and k are unit vectors along x, y and z. 10

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1

$$T = \frac{2\pi}{\omega}$$

$$r = R + h$$

$$\frac{GM}{r^2} = \omega^2 r$$

$$\frac{GM}{r^3} = \omega^2$$

$$\omega = \sqrt{\frac{GM}{r^3}}$$

a) (i) Show that the differential cross-section can be expressed as

$$\sigma(\theta) = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right| \quad 10$$

(ii) Obtain an expression for Rutherford Scattering cross section and interpret the result. 15

(b) State and prove Kepler's third law of planetary motion. 10

(c) A particle of mass m is observed to move in a spiral orbit given by the equation $r = C\theta$, where C is a constant. Find the force law if particle is moving in a central force field. Also find $\theta(t)$. 15

3. (a) For a free-falling body from the height ' h ' on the surface of the earth in the northern hemisphere with a latitude θ , show that the deviation of the body towards the east at the final stage is given by $\frac{1}{3} \omega \cos\theta \left(\sqrt[3]{\frac{8h^3}{g}} \right)$, where ω is the angular velocity of the earth and ' g ' is the acceleration due to gravity. 15

(b) Find the center of mass of

(i) A non-uniform rod of length l and mass per unit length as $\lambda = A \cos(\pi x/2l)$ where x is position along the rod $0 \leq x \leq l$? 10

(ii) A water molecule H_2O consisting of central oxygen atom bound to two hydrogen atoms. The two hydrogen-oxygen bonds subtend an angle of 104.5° and each bond has a length of 0.097 nm . 10

(c) (i) Show that Kepler second law of planetary motion is a consequence of conservation of angular momentum. 5

(ii) Prove that the time taken by earth to travel over half of its orbit separated by the minor axis remote from the sun is two days more than half a year. Given the time period of earth is 365 days and eccentricity of the orbit is $1/60$. 10

Q4. (a) What do you understand by Fictitious Forces? Derive the expression for Coriolis force and centrifugal force in a rotating coordinate system. 15

(b) (i) Show that in Centre of mass frame of reference the speed of two colliding particle remains the same but velocity vector changes. 10

(ii) A proton makes a head-on collision with an unknown particle at rest and rebounds straight back with $4/9^{\text{th}}$ of its initial Kinetic Energy. Assuming elastic collision find the ratio of the mass of unknown particle to the mass of proton. 10

(c) A right-angled triangle prism of mass ' M ' with angle θ rests on a horizontal surface with its base on the ground. A small block of mass ' m ' is placed on the prism. Find the acceleration of the prism and the block assuming all surfaces to be frictionless. 15

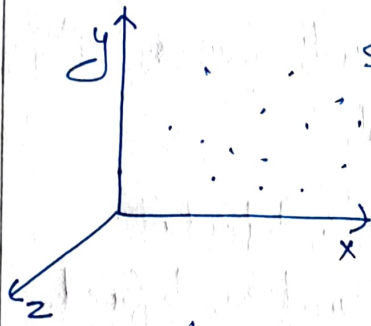
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Mechanics of Particles

CLASS TEST

Q1 (a)



System of N Particles

$$\vec{L} = \sum_{i=1}^N \vec{r}_i \times \vec{p}_i$$

Differentiating w.r.t t ,

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^N \vec{r}_i \times \vec{F}_i \quad \text{--- (1) (Newton's 2nd law)}$$

$$\frac{d\vec{p}_i}{dt} = \vec{F}_i$$

$$\vec{F}_i = \vec{F}_i^{\text{external}} + \sum_{j \neq i} \vec{F}_{ij} \quad \text{--- (2)}$$

Substituting $\vec{F}_i$ from (2) to (1), $\Rightarrow$

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^N \vec{r}_i \times (\vec{F}_i^{\text{ext}} + \sum_{j \neq i} \vec{F}_{ij})$$

$$\frac{d\vec{L}}{dt} = \sum_{i=1}^N \vec{r}_i \times \vec{F}_i^{\text{ext}} + \sum_{i=1}^N \sum_{\substack{j \neq i \\ j=1}}^N \vec{r}_i \times \vec{F}_{ij}$$

Net external torque on the system $= 0 \Rightarrow$

$$\sum_{i=1}^N \vec{r}_i \times \vec{F}_i^{\text{ext}} = 0, \therefore$$

$$\frac{d\vec{L}}{dt} = \sum_{j \neq i} \vec{r}_i \times \vec{F}_{ij}$$

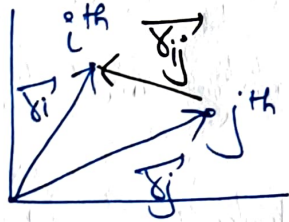
$$= \frac{1}{2} \sum \sum (\vec{r}_i \times \vec{F}_{ij} + \vec{r}_j \times \vec{F}_{ji})$$

Since $\vec{F}_{ij} = -\vec{F}_{ji}$ (Newton's 3rd law),

the above equation becomes,

$$\frac{d\vec{L}}{dt} = \frac{1}{2} \sum_{j \neq i} (\vec{r}_i - \vec{r}_j) \times \vec{F}_{ij}$$

But Newton's 3rd law doesn't ensure that the force is central but,



An angle dependent force cannot be defined using a single $\vec{r}_{ij}$

$$\Rightarrow \vec{r}_{ij} \text{ is along } \vec{F}_{ij} \Rightarrow$$

$$\boxed{\frac{d\vec{L}}{dt} = 0} \quad \therefore \boxed{|\vec{L}| = \text{Constant}}$$

211 (d)

Let W_{NC} = Work done by a ^{Non} conservative force.

W_C = Work done by conservative force.

$$W_{net} = W_{NC} + W_C \quad \text{--- (1)}$$

For conservative forces,

$$W_C = \int_{x_a}^{x_b} \vec{F} \cdot d\vec{x} = U(a) - U(b) \quad \text{--- (2)}$$

From work energy theorem,

$$W_{net} = \Delta K \quad \text{i.e.}$$

$$W_{net} = K_b - K_a \quad \text{--- (3)}$$

Substituting (2) & (3) in (1) $\Rightarrow$

$$W_{net} = W_{NC} + W_C$$

$$\Delta K = W_{NC} + U(a) - U(b)$$

$$K_b - K_a = W_{NC} + U(a) - U(b)$$

$$\therefore W_{NC} = E(b) - E(a)$$

where $E(b) = K_b + U(b)$ Mech. Energy of particle at b.

$E(a) = K_a + U(a)$ Mech. Energy of particle at a.

$$\therefore \boxed{W_{NC} = E(b) - E(a)}$$

21) (c)

Conservative Force - The work done by such forces is path independent, & work done along a closed path is

0 i.e.
$$\oint \vec{F}_{\text{conservative}} \cdot d\vec{x} = 0$$

The Mechanical energy for such forces remains conserved i.e

$$K_a + U(a) = K_b + U(b) \quad \text{--- ①}$$

From work-energy theorem,

$$W_{\text{net}} = \Delta K$$
$$\int_{x_a}^{x_b} \vec{F} \cdot d\vec{x} = K_b - K_a$$

From ①, $K_b - K_a = U(a) - U(b)$

$$\therefore \int_{x_a}^{x_b} \vec{F} \cdot d\vec{x} = U(b) - U(a)$$

$\therefore$ If $x_a = x_b$ i.e. $\oint_{x_a}^{x_a} \vec{F} \cdot d\vec{x} = 0$

only for conservative forces.

Also,
$$\vec{F} = -\vec{\nabla} U$$

Given,
$$\vec{F} = (y^2 z^3 - 6xz^2)\hat{i} + 2xyz^3\hat{j} + (3xy^2z^2 - 6x^2z)\hat{k}$$

For a force to be conservative,

$$\text{Curl } \vec{F} = 0 \quad \text{i.e.}$$

$$\text{Curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2z^3 - 6xz^2 & 2xyz^3 & 3xy^2z^2 - 6x^2z \end{vmatrix}$$

$$= \hat{i} [6yz^2 - 6yz^2] - \hat{j} [3y^2z^2 - 12xz - (3y^2z^2 - 12xz)]$$

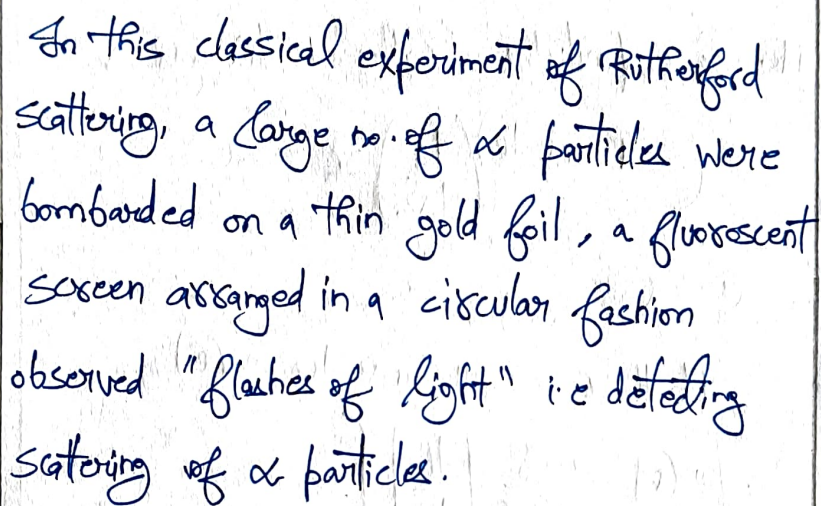
$$+ \hat{k} [2yz^3 - 2yz^3 + 0]$$

$$= 0$$

Hence the force field is conservative in nature.

(i)

॥



$b =$ Impact parameter.

For the scattered α particles, all the particles crossing the circular strip of area $2\pi b db$ must scatter through the $d\Omega$ solid angle subtended by circular strip of fluorescent screen.

Therefore,

$$\text{No. of particles crossing circular strip} \\ = (2\pi b db) I \quad - (1)$$

Solid angle subtended by the strip at the centre i.e. at Nucleus = $\frac{\text{perpendicular area}}{r^2}$

$$\Omega = \frac{(2\pi r \sin\theta)(r d\theta)}{r^2} = 2\pi \sin\theta d\theta$$

$$\therefore \text{No. of } \alpha \text{ particles scattered through } \Omega \\ = (2\pi \sin\theta d\theta) \left(\frac{d\sigma}{d\Omega} \right) I \quad - (2)$$

$$\therefore (1) = (2)$$

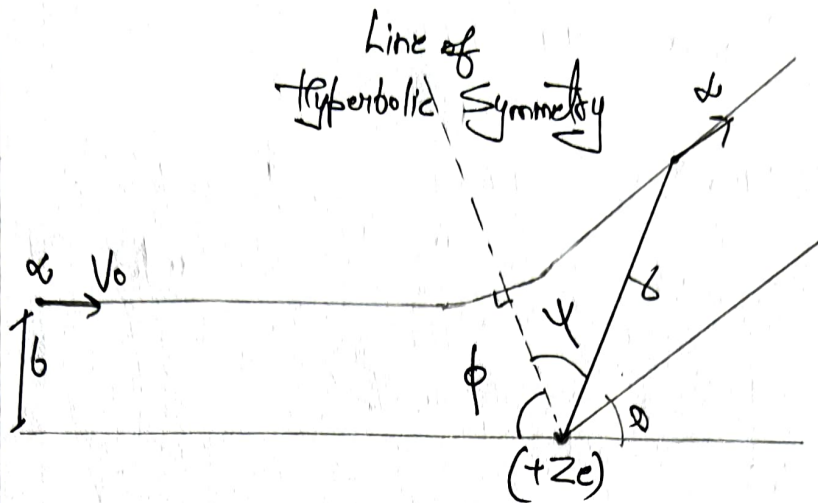
$$\Rightarrow 2\pi b db I = 2\pi \sin\theta d\theta \sigma(\theta) I$$

$$\therefore \boxed{\sigma(\theta) = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|}$$

ii)

Assumptions in this experiment

- a) No Quantum effects.
- b) Ignoring the recoil of nucleus.
- c) Interaction to be purely Coulombic.
- d) Non-relativistic velocity.



Consider an scattered α particle through angle θ & located at polar coordinates (r, ψ) at any time t' .

$$\Rightarrow 2\phi + \theta = \pi \Rightarrow \boxed{\phi = \frac{\pi - \theta}{2}} \quad -①$$

Since the Coulombic force is central, Angular mom. is conserved $\therefore$

$$\boxed{L = mV_0 b} \quad -②$$

For central force motion, the ~~er~~ locus of the particle can be shown by,

$$\boxed{\frac{r}{\gamma} = 1 + e \cos \psi} \quad ; \text{ where } e = \sqrt{1 + \frac{2EJ^2}{\mu K^2}} \quad -③$$

As $\psi \rightarrow \alpha$, $\psi \rightarrow \phi$

$\therefore$ Eq. ③ becomes, $0 = 1 + e \cos \phi$

$$\therefore \cos \phi = -\frac{1}{e} \quad \text{or} \quad \cos\left(\frac{\pi - \theta}{2}\right) = -\frac{1}{e}$$

$$\sin \frac{\theta}{2} = \frac{1}{e} \quad \therefore \sec \frac{\theta}{2} = -e$$

$$\text{or} \quad \sec^2 \frac{\theta}{2} = e^2$$

$$\sec^2 \frac{\theta}{2} = 1 + \frac{2EL^2}{mK^2}$$

$$1 + \sec^2 \frac{\theta}{2} = 1 + \frac{2EL^2}{mK^2} ; L = mv_0 b$$

$$E = \frac{1}{2} m v_0^2$$

$$\sec^2 \frac{\theta}{2} = \frac{2}{mK^2} \cdot E \cdot m^2 v_0^2 b^2$$

$$\propto \sec^2 \frac{\theta}{2} = \left(\frac{2E}{K} \right)^2 b^2 \Rightarrow \boxed{b = \frac{K \sec \frac{\theta}{2}}{2E}}$$

(4)

$$\frac{db}{d\theta} = -\frac{K}{2E} \sec^2 \frac{\theta}{2} \cdot \frac{1}{2}$$

$$\text{Now, } \sigma(\theta) = \frac{b}{\sin \theta} \left| \frac{db}{d\theta} \right|$$

$$\Rightarrow \sigma(\theta) = \frac{b}{\sin \theta} \frac{K}{2E} \left(\sec^2 \frac{\theta}{2} \right) \cdot \frac{1}{2}$$

$$\sigma(\theta) = \frac{b}{4} \left(\frac{Zze^2}{4\pi\epsilon_0} \right) \cdot \frac{1}{\frac{1}{2} m v_0^2} \frac{\sec^2 \frac{\theta}{2}}{\sin \theta}$$

$$= \frac{Zze^2}{8\pi\epsilon_0 m v_0^2} \cdot \frac{K}{2E} \frac{\sec \frac{\theta}{2} \sec^2 \frac{\theta}{2}}{\sin \theta}$$

$$\boxed{\sigma(\theta) = \left(\frac{K^2}{8E^2} \right) \sec^4 \frac{\theta}{2}} ; \boxed{K = \frac{Zze^2}{4\pi\epsilon_0}}$$

$$\boxed{E = \frac{1}{2} m v_0^2}$$

Consequences of Above expression

① The α particles with large Kinetic Energy are only deflected through a small θ .

② For α particles with large K.E bombarded on a light nucleus (with just 10-15 protons), the α particles reaches very close to the atomic nucleus due to weak electrostatic repulsion, therefore helped in predicting the radius of nucleus,

$$R = R_0 A^{\frac{1}{3}}$$

where $R_0 = 1.4 \text{ fm}$

$A = \text{protons} + \text{neutrons}$

(Mass no.)

$$\textcircled{3} \quad \sigma(\theta) \propto \frac{1}{\sin^4 \frac{\theta}{2}} \quad \Rightarrow \quad \text{As } \theta \uparrow \text{ es, } \sigma(\theta) \downarrow \text{ es.}$$

Q2) (b)

Kepler's 3rd law of planetary motion.

To prove $T^2 \propto a^3$.

From Kepler's 2nd law, $\frac{dA}{dt} = \frac{L}{2m}$;

(a simple consequence of conservation of angular momentum due to central nature of force).

$$\therefore T = ? \quad \frac{\pi ab}{T} = \frac{L}{2m}$$

$$\Rightarrow \boxed{T = \frac{2\pi ab}{L}} \quad \text{--- (1)}$$

For central force, $\frac{r}{r} = 1 + e \cos(\theta - \theta_0)$; $\boxed{l = \frac{J^2}{mK}}$ (4)

$$r_{\max} = \frac{l}{1-e} ; r_{\min} = \frac{l}{1+e}$$

$$r_{\max} + r_{\min} = 2a = \frac{l}{1-e} + \frac{l}{1+e}$$

$$2a = \frac{l[1+e+1-e]}{1-e^2}$$

$$\Rightarrow 2a = \frac{2l}{1-e^2} \Rightarrow \boxed{l = a(1-e^2)} \quad \text{--- (2)}$$

For ellipse, $\frac{b}{a} = \sqrt{1-e^2} = \sqrt{\frac{l}{a}}$

$$\therefore \boxed{b = \sqrt{al}} \quad \text{--- (3)}$$

$$T = \frac{2\pi ma}{L} \cdot \sqrt{al} \Rightarrow T = \frac{2\pi m \sqrt{a^3}}{L} \quad \text{--- (4)}$$

$$T = \frac{2\pi m}{K \sqrt{mk}} \cdot a^{\frac{3}{2}}$$

$$T = \frac{2\pi m}{\sqrt{mk}} a^{\frac{3}{2}}$$

$$T^2 = \frac{4\pi^2 m}{K} a^3$$

$$K = GMm$$

$$\therefore T^2 \propto a^3$$

Kepler's 3rd law of Planetary motion.

Q.2 (c)

The polar trajectory of particle = $c\theta$,
 $c = \text{Constant}$.

Provided that the particle is moving in
Central force field,

$$r = c\theta,$$

For central force motion,

$$\boxed{\frac{d^2u}{d\theta^2} + u = -mF\left(\frac{1}{u}\right)} \quad \begin{array}{l} J = \text{Conserved angular} \\ \text{momentum,} \\ u = \frac{1}{r} \end{array}$$

$$\therefore \frac{1}{u} = c\theta \quad \text{or} \quad u = \frac{1}{c\theta}$$

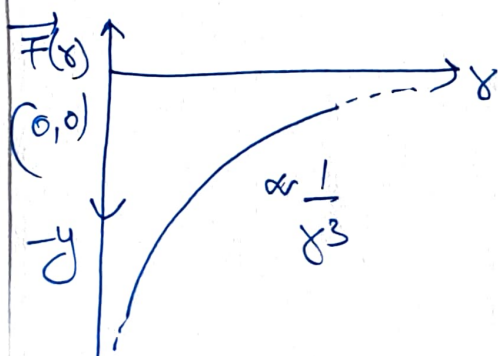
$$\text{or} \quad \frac{du}{d\theta} = -\frac{1}{c\theta^2} \quad ; \quad \frac{d^2u}{d\theta^2} = \frac{1}{c\theta} = u$$

$$\therefore u + u = \frac{-m}{J^2 u^2} \cdot F\left(\frac{1}{u}\right)$$

$$\text{or} \quad F\left(\frac{1}{u}\right) = -\frac{J^2 u^2}{m} (2u)$$

$$F\left(\frac{1}{u}\right) = -\frac{2J^2 u^3}{m} \quad ; \quad \text{Since } \frac{1}{u} = r$$

$$\Rightarrow \boxed{\vec{F}(r) = -\frac{2J^2}{m r^3} \hat{r}} \quad \text{Attractive force.}$$



Find $\theta(t) = ?$

Since $J = m r^2 \dot{\theta}$

$r = c\theta$ in above eqⁿ,

$$J = m(c^2 \theta^2) \frac{d\theta}{dt}$$

$$\therefore \int_0^t \frac{J}{mc^2} dt = \int_{\theta_0}^{\theta} \theta^2 d\theta$$

Assuming $\theta_0 = 0$ at $t = 0$

$$\Rightarrow \frac{Jt}{mc^2} = \frac{\theta^3}{3}$$

$$\therefore \theta = \left(\frac{3Jt}{mc^2} \right)^{\frac{1}{3}}$$

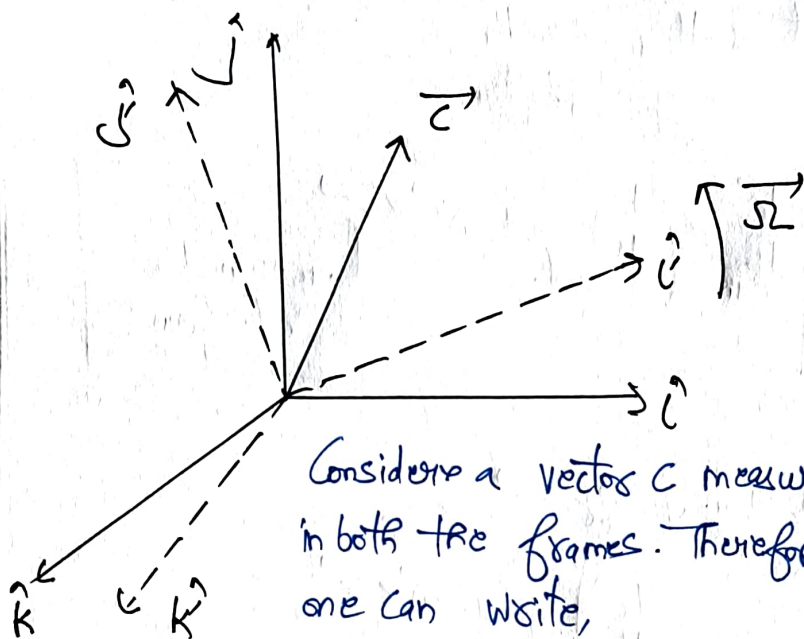
$$\theta \propto t^{\frac{1}{3}}$$

(As the Angular momentum J is already conserved for central force field).

Q4) (a)

Fictitious Forces — These are not actual physical forces acting on an object but arise due to the accelerated frames.

Consider a coordinate system $(\hat{i}, \hat{j}, \hat{k})$ & a rotating coordinate system $(\hat{i}', \hat{j}', \hat{k}')$ with respect to $\hat{i}, \hat{j}, \hat{k}$ with angular velocity $\vec{\omega}$.



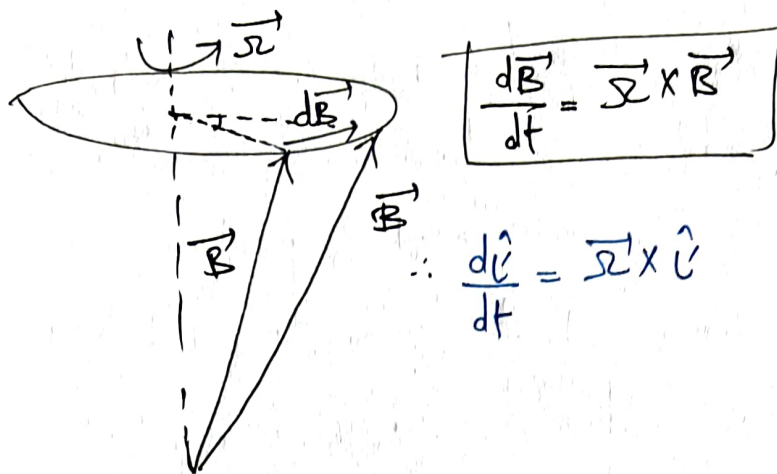
Consider a vector $\vec{C}$ measured in both the frames. Therefore one can write,

$$\vec{C} = C_x \hat{i} + C_y \hat{j} + C_z \hat{k} = C'_x \hat{i}' + C'_y \hat{j}' + C'_z \hat{k}' \quad \text{--- (1)}$$

$$\therefore \frac{d\vec{C}}{dt} = \frac{d}{dt} (C'_x \hat{i}' + C'_y \hat{j}' + C'_z \hat{k}')$$

$$\begin{aligned} \frac{d\vec{C}}{dt} &= \frac{dC'_x}{dt} \hat{i}' + C'_x \frac{d\hat{i}'}{dt} + \frac{dC'_y}{dt} \hat{j}' + C'_y \frac{d\hat{j}'}{dt} + \frac{dC'_z}{dt} \hat{k}' + C'_z \frac{d\hat{k}'}{dt} \\ &\quad + \frac{d\hat{k}'}{dt} \cdot C'_z \end{aligned}$$

For any vector $\vec{B}$ rotating with angular velocity $\vec{\omega}$;



$$\therefore \frac{d\vec{c}}{dt} = \left(\frac{d\vec{c}}{dt} \right)_{\text{measured in inertial frame}} + \vec{\omega} \times (c'_x \hat{i} + c'_y \hat{j} + c'_z \hat{k})$$

rotational

$$\text{or } \boxed{\frac{d\vec{c}}{dt}_{\text{in}} = \left(\frac{d\vec{c}}{dt} \right)_{\text{rot.}} + \vec{\omega} \times \vec{c}} \quad \text{measured in rot. frame}$$

-①

Now, Using ① for $\vec{r}$ $\Rightarrow$

$$\left(\frac{d\vec{r}}{dt} \right)_{\text{in}} = \left(\frac{d\vec{r}}{dt} \right)_{\text{rot.}} + \vec{\omega} \times \vec{r}$$

$$\text{or } \boxed{\vec{V}_{\text{in}} = \vec{V}_{\text{rot.}} + \vec{\omega} \times \vec{r}} \quad -②$$

Now using ① for $\vec{V}_{\text{in}}$,

$$\left(\frac{d\vec{V}_{\text{in}}}{dt} \right)_{\text{in}} = \left(\frac{d\vec{V}_{\text{in}}}{dt} \right)_{\text{rot.}} + \vec{\omega} \times \vec{V}_{\text{in}}$$

Substituting the value of $\vec{V}_{\text{in}}$ from ② in above $\Rightarrow$

$$\vec{a}_{\text{in}} = \frac{d}{dt} (\vec{V}_{\text{rot.}} + \vec{\omega} \times \vec{r}) + \vec{\omega} \times (\vec{V}_{\text{rot.}} + \vec{\omega} \times \vec{r})$$

$$\vec{a}_{\text{in}} = \vec{a}_{\text{rot.}} + \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \vec{V}_{\text{rot.}} + \vec{\omega} \times \vec{V}_{\text{rot.}} + \vec{\omega} \times (\vec{\omega} \times \vec{r})$$

Multiply the above eqⁿ with 'm', rearrange,

$$m\vec{a}_{\text{rot}} = m\vec{a}_{\text{in}} - 2m(\vec{\omega} \times \vec{v}_{\text{rot}}) - m\left[\frac{d\vec{\omega}}{dt} \times \vec{r}\right] - m\vec{\omega} \times (\vec{\omega} \times \vec{r})$$

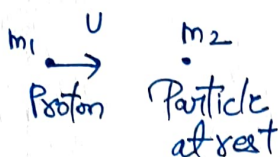
$$\therefore \boxed{\vec{F}_{\text{Coriolis}} = -2m(\vec{\omega} \times \vec{v}_{\text{rot}})}$$

$$\boxed{\vec{F}_{\text{Euler}} = -m\left(\frac{d\vec{\omega}}{dt} \times \vec{r}\right) = -m(\vec{\alpha} \times \vec{r})}$$

$$\boxed{\vec{F}_{\text{centrifugal}} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r})}$$

24/6)
(ii)

Before
Collision



After collision.



Since the collision is elastic, the Energy remains conserved,

$$\frac{1}{2} m_1 U^2 = \frac{1}{2} m_1 V_1^2 + \frac{1}{2} m_2 V_2^2 -$$

$$\text{or } m_1 U^2 = m_1 V_1^2 + m_2 V_2^2 - \text{①}$$

$\vec{F}_{\text{net}}^{\text{ext}} = 0 \Rightarrow$ Linear momentum remains conserved, $\therefore$

$$m_1 U = m_1 V_1 + m_2 V_2 - \text{②}$$

$$\text{Let } \frac{m_2}{m_1} = \gamma \quad \therefore \quad \boxed{U = V_1 + \gamma V_2}$$

$$\text{Also, } \cancel{\frac{1}{2} m_1 V_1^2} = \frac{4}{9} \left(\cancel{\frac{1}{2} m_1 U^2} \right)$$

$$\therefore \quad \boxed{V_1 = \frac{2}{3} U}$$

$$\therefore \quad U = \frac{2}{3} U + \gamma V_2 \Rightarrow \frac{U}{3} = \gamma V_2 - \text{③}$$

$$\text{Also, ① becomes, } U^2 = V_1^2 + \gamma V_2^2$$

$$U^2 = \frac{4}{9} U^2 + \gamma V_2^2 \quad \therefore$$

Substitute, $V_2 = \frac{U}{3\gamma}$ in above eqⁿ,

$$U^2 - \frac{4}{9} U^2 = \gamma \frac{U^2}{9\gamma^2} \Rightarrow \frac{5}{9} = \frac{1}{\gamma}$$

$$\Rightarrow \quad \boxed{\gamma = \frac{1}{5}} \quad \text{or} \quad \boxed{\frac{m_2}{m_1} = \frac{1}{5}}$$